L0 - Course Preliminaries

Topics covered

- ✧ Course basic information
- ✧ The needs and offers from the most prominent IT companies
- ✧ The challenges of industrial internship, final year projects and the first job

Course basic information

Lecturer

Dr. YEUNG Check Yu, Leo

- Honorary Lecturer, PhD and BEng in Dept. of Computer Science at HKU
- Managing Director and Founder at Una Technologies Limited
  - Founded in 2017 in Hong Kong Science Park
  - Lead R&D team to develop in-house EdTech platform and online solutions for clients' tender projects
- Awards and endorsements:

Microsoft in Education Global Training Partner

Gold Hong Kong ICT Awards 2019 香港資訊及通訊科技獎 Social Impact

Bronze Award 銅獎 Hong Kong ICT Awards 2025 香港資訊及通訊科技獎

TECH 2025 SINGULARITY Smart City AI Content 2025 Second Place

Reimagine EDUCATION Challenge First Runner-up Award Winner

CERTIPORT A PEARSON VUE BUSINESS AUTHORIZED TESTING CENTER

ITC ESS

HKTDC 香港貿發局





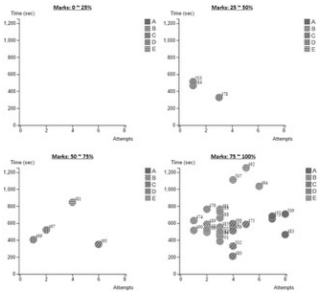


Leo's previous projects

✧ Commercial, industrial and R&D projects, involving EdTech, AI, IoT, …



1) Una Platform: Real-time interactive AI education platform (www.una.study)
- 7+ years continuous development
- Served 400+ schools and 35,000+ students



2) SkyApp: Collaborative learning system for Maths
- Opensource
- Research



3a) Making Smart Choices 2: Sexuality Education Experiential Learning Game
3b) Hall Mission: The Gamified Q&A System
- Game
- Incremental development



4) Others: Flash games revamp, online stores, CRM, company websites

Course learning outcomes

- Software engineering **foundation**
  - Able to demonstrate an understanding of the **software development process** and activities including the **analysis, design, implementation, testing** and deployment of contemporary software systems.
- Software engineering **methodology**
  - Able to **apply** the software process and process activities to real-world software development.
- Software engineering **tools and techniques**
  - Able to use industry standard tools and techniques for software development.
- Software engineering **practice**
  - Able to **plan**, **work** and **contribute effectively** as a member of self-organizing software development **teams**.

Summarized all learning outcomes into one goal!



Novice
programmer



Professional
software engineer

(or project manager, client)

2026-27 Assessments at a Glance

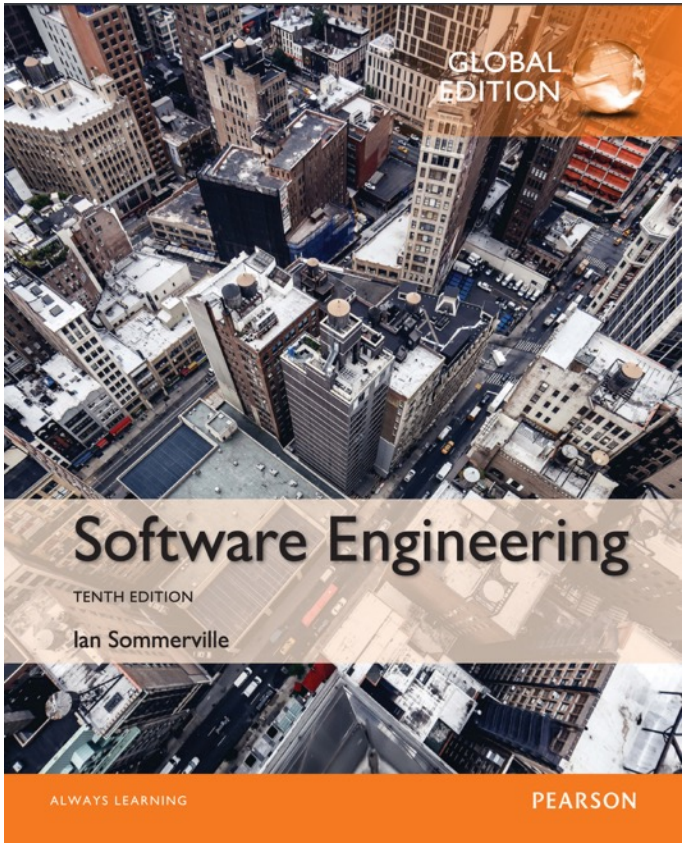
- ✧ No quiz this year.
- ✧ Workshops - 15%: 7 assessed workshop tasks (W3, W4, W5, W6, W7, W8, W9); best 5 count, 3% each; individual work, submitted within 7 days of the workshop.
- ✧ Group Project - 35%: build a WORKING web application for a client (played by TA); all work on GitHub. A working product carries the biggest weight - alongside requirements, engineering process and individual contribution.
- ✧ Exam - 50%: covers lectures L1 to L10 and all workshops.

How This Course Uses AI

- AI tools are allowed and expected - in every workshop and throughout the project.
- Free for EVERY student - nobody has to pay to complete this course: GitHub Copilot via GitHub Education, set up in W1; Cloudflare's free tier, from W2.
- The rule that never changes: you must be **able to explain every line you submit**.
- AI use is declared in your Individual Contribution & AI-Usage Report (one per student) - graded on honesty, not on how much AI you used.
- In industry these tools are paid products - budget for them in real projects.
- You may use your own PAID AI for workshops and the project - but our marking is designed so paid tools do not easily buy higher marks.

AI-use policy follows HKU Teaching & Learning - GenAI and plagiarism

Main Textbook



Software Engineering, 10th Edition, Ian Sommerville, Pearson (2016)

Lectures at a Glance

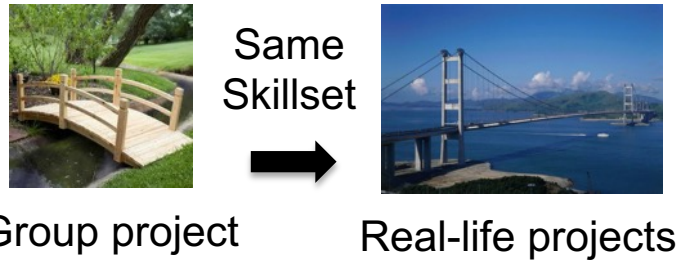
Lecture	Topic
L1	Introduction
L2	Requirements Engineering
L3	Modeling to Communicate (UML-lite)
L4	UX and Prototyping
L5	Design: Architecture, Patterns & SOLID
L6	Implementation with AI
L7	Testing and QA
L8	Evolution and Technical Debt
L9	Software Processes
L10	Agile Software Development
L11	Latest Trends

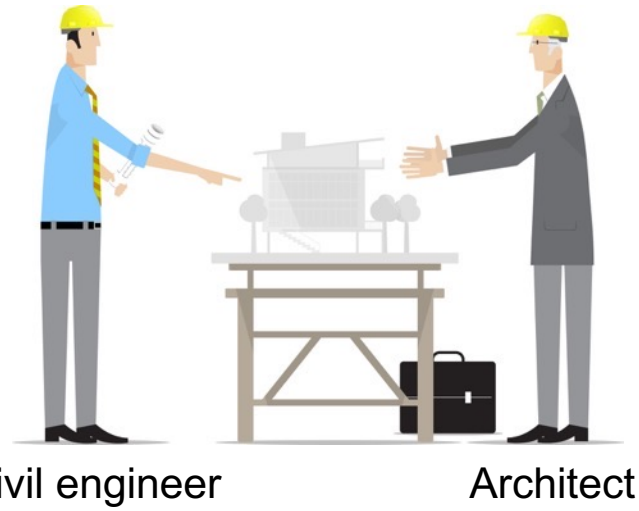
Group project

- Group size: 5 - 6 students (self-group to max 3 students)
- A specific project assigned by the client (simulated by the TAs)
- The project runs in phases, each with a deadline and deliverables:
  - *Group formation*
  - Client meeting #1 - Requirements elicitation
  - Deliverables #1: User requirements
  - Deliverables #2: MVP v1.0 - a working web app + requirements
  - Client meeting #2 - live demo & presentation
  - Deliverables #3: Final v1.1 - change + tests + coverage
  - Individual Contribution & AI-Usage Report (one per student)

Group project FAQs

- Why do we have group work?
  - Ans: Most real SE projects are done in **teams**.
- Do we need to code in group project?
  - Ans: Yes. "Vibe coding" with AI is allowed and expected - but you must be able to explain every line you submit.
- What should I do if free-rider(s) exist in my team?
  - Ans: In your **Individual Contribution & AI-Usage Report** - includes peer evaluation; a free-rider can score 0.





Hands-up survey

✧ Do you have any experience using the following software engineering tools? (Raise your hand if Yes)

1. GitHub (version control & collaboration)
2. GitHub Copilot (AI-assisted coding)
3. Figma (UI/UX design)
4. Python unittest (unit testing)
5. Cloudflare (cloud deployment)



Workshops

Workshop	Topic	Assessed
W1	Git, AI & Tools Setup	-
W2	User Stories & Acceptance Criteria with AI	-
W3	UML — Drawing It, and Checking the AI	✓
W4	UX/UI — Designing It With AI, and Owning the Decisions	✓
W5	From Design to Deployed — Building Someone Else's Design	✓
W6	Plan Review — Making the Agent Show Its Working	✓
W7	Testing I — unittest and Assertions	✓
W8	Testing II — Coverage, and What It Misses	✓
W9	Evolution and Hotfix — one release, one change, one emergency	✓

Assessed workshops (W3-W9): seven tasks, best 5 count, 3% each; individual work, submitted within 7 days.

Where to find information and help

- ✧ **Moodle course page**:
  - https://moodle.hku.hk/course/view.php?id=139763
  - Including Details of **Schedule**, **Consultation**, etc
- ✧ Important updates will be announced in the **News announcements.**
- ✧ As usual, you can post questions in the **Course Forum**.
  - Guideline #1: When you post a question about course materials, post your best guess with answers, so we can understand your questions.
  - Guideline #2: For assessments, **don't** post your answers!

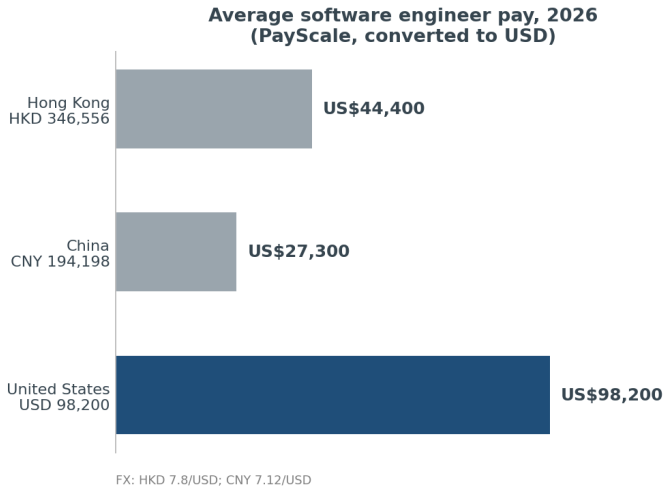
Course FAQs

- Can I join this course?
  - **Quota is limited** and handled centrally by CS General Office due to departmental policy
  - Contact enroll@cs.hku.hk and outline your situation
- Any lesson video recordings?
  - Referring to the HKU note on T&L arrangement in 2022, teachers are required to deliver **face-to-face** lessons.
  - All lessons will be **recorded**. Only open **7-day access** to students who submit requests with **valid proof**, e.g. medical cert.
  - To request **video recordings**, send an email to the Lecturer together with i) student name, ii) student ID, iii) date of the absented lesson(s), iv) reason, and iv) valid proof.

The needs and offers from the most prominent IT companies

Career path

- Software Engineer **Avg. Salary (2026)**
  - in Hong Kong: **HKD 346,556** / year
  - in US: **USD 98,200** / year
  - In China: **CNY 194,198** / year
- **FAANG** (Facebook, Amazon, Apple, Netflix, and Google) / **MATANA** (+ TESLA), are Tier 1 tech companies and fast-growing tech startups are known for paying high salaries to software engineers.
- China AI giant DeepSeek: **up to CNY 1.54M** / year (Dim Sum Daily, 2025)



(Refs: HK - PayScale 2026; US - PayScale 2026; China - PayScale 2026; DeepSeek - Dim Sum Daily, Feb 2025)

10 Highest-Paying IT Jobs in Hong Kong

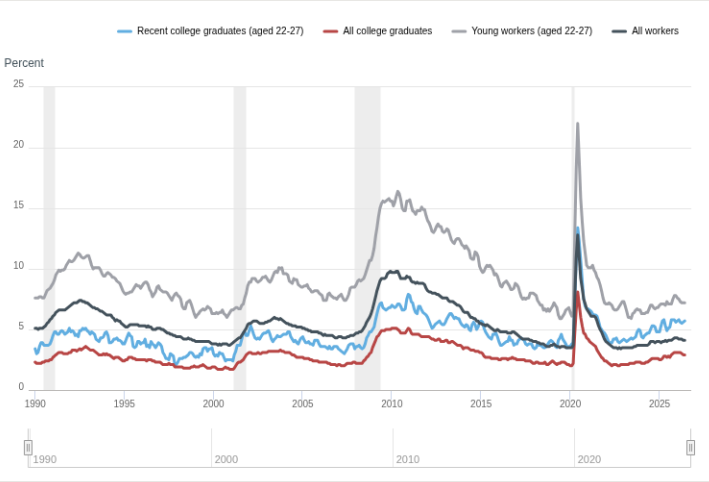
- Chief Technology Officer (CTO) - HK$83,000-166,000/month
- Solutions Architect - about HK$63,000/month
- IT Manager - about HK$50,000/month
- System Analyst - HK$47,000-103,000/month
- Data Manager - about HK$43,000/month
- Network Engineer - about HK$41,000/month
- Database Administrator - about HK$41,000/month
- Cyber Security Analyst - about HK$37,000/month
- Data Scientist - about HK$34,000/month
- Software Engineer - about HK$33,000/month - the entry point to every role above



(Reference: SVA Recruitment, 2026)

The 2026 Job Market - the Honest Picture

- **Getting your first tech job is hard right now - I will not pretend otherwise.**
  - Same story in the US: fresh-graduate unemployment in Computer Science is about **6.1%** - among the highest of all majors (NY Fed, 2023 data - via Newsweek)
  - Hong Kong, same story in software jobs: graduate openings for conventional **IT-programming roles fell over 60%** year-on-year (JIJIS Q1 2026, via The Standard)
- Graduate vacancies posted with the eight public universities: **30,798** in 2025 - the fewest in five years, **down 55%** from 68,728 in 2024.
  - Source: JIJIS, via Macao News
- Why: post-pandemic over-hiring correction + **AI automating routine junior tasks**.



(Unemployment: recent grads vs all workers - NY Fed)

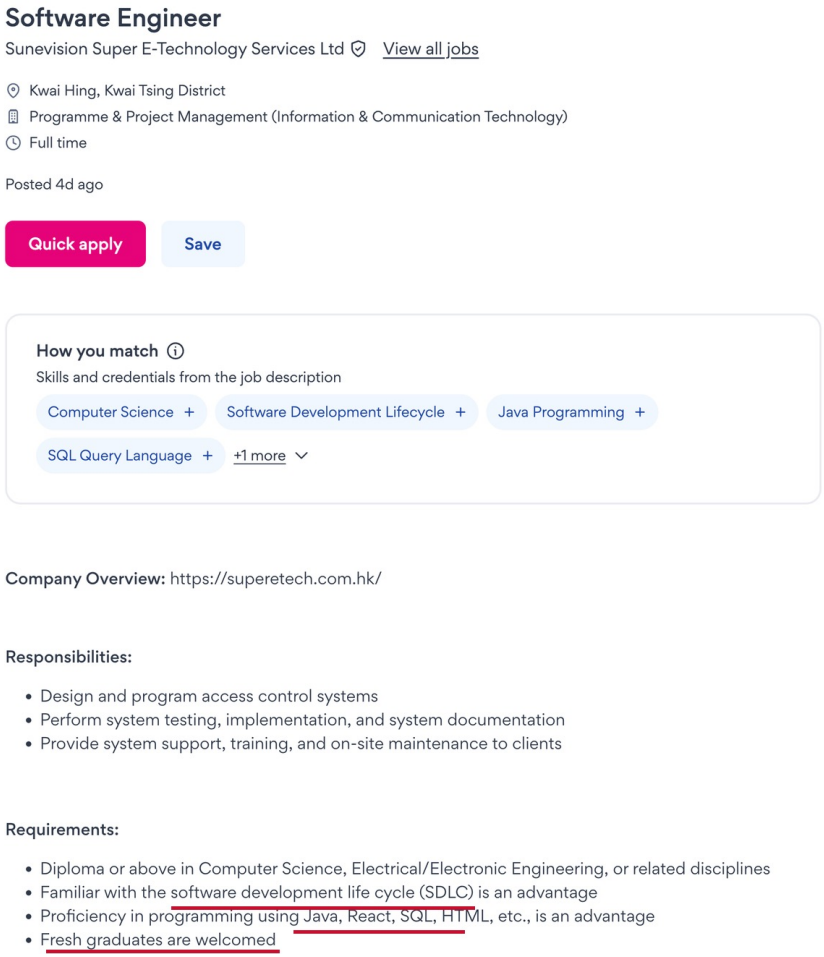
The Same Data, Read Carefully - Skill Still Sells

- The US still projects about **129,200** software-developer openings a year through 2034 - **15% growth**, much faster than average; median pay **US$133,080**.
  - Source: US Bureau of Labor Statistics, Occupational Outlook Handbook
- Talent shortage is the **#1 obstacle** to IT investment in Hong Kong - yet **over half** of organisations still plan to INCREASE IT spending in 2026 (top skills: AI/ML 33.6%, cybersecurity 31.9%, data 12.9%).
  - Source: Hong Kong Computer Society, 2026 IT Industry Trend Survey (press release, 14 Jun 2026)
- **One in four** HK companies cannot fill AI-application roles; the median graduate starting salary still **rose ~2% to HK$20,000/month** (Q1 2026).
  - Sources: ManpowerGroup survey, Dec 2025 - via China Daily HK; JIJIS Q1 2026 pay data - via The Standard, May 2026
- Generative-AI job postings in the US grew about **170%** in one year - AI is moving demand up the stack, not removing it.
  - Source: Indeed postings analysis, Jan 2025 - via CIO Dive

Our Response: Become the Engineer AI Cannot Replace

- The junior tasks AI automates are exactly the tasks this course trains you to **SUPERVISE**: requirements, design, testing, review.
- This course is rebuilt for that: you will **ship a real product, with AI, under engineering discipline** - stories, UML, tests, CI, deployment.
- Your portfolio after 13 weeks: a deployed web app + a GitHub history + a test suite with CI - **evidence, not claims**.
- **The market is tough on average engineers and eager for excellent ones. My goal: you leave this course on the excellent side of that line.**
  - All sources on the previous two slides are public - verify them yourself; that habit is also part of being an engineer.

Latest Job Ads in Hong Kong - JobsDB



Software Engineer

Sunevision Super E-Technology Services Ltd View all jobs

Kwai Hing, Kwai Tsing District
Programme & Project Management (Information & Communication Technology)
Full time

Posted 4d ago

Quick apply Save

How you match
Skills and credentials from the job description
Computer Science + Software Development Lifecycle + Java Programming +
SQL Query Language + +1 more

Company Overview: https://superetech.com.hk/

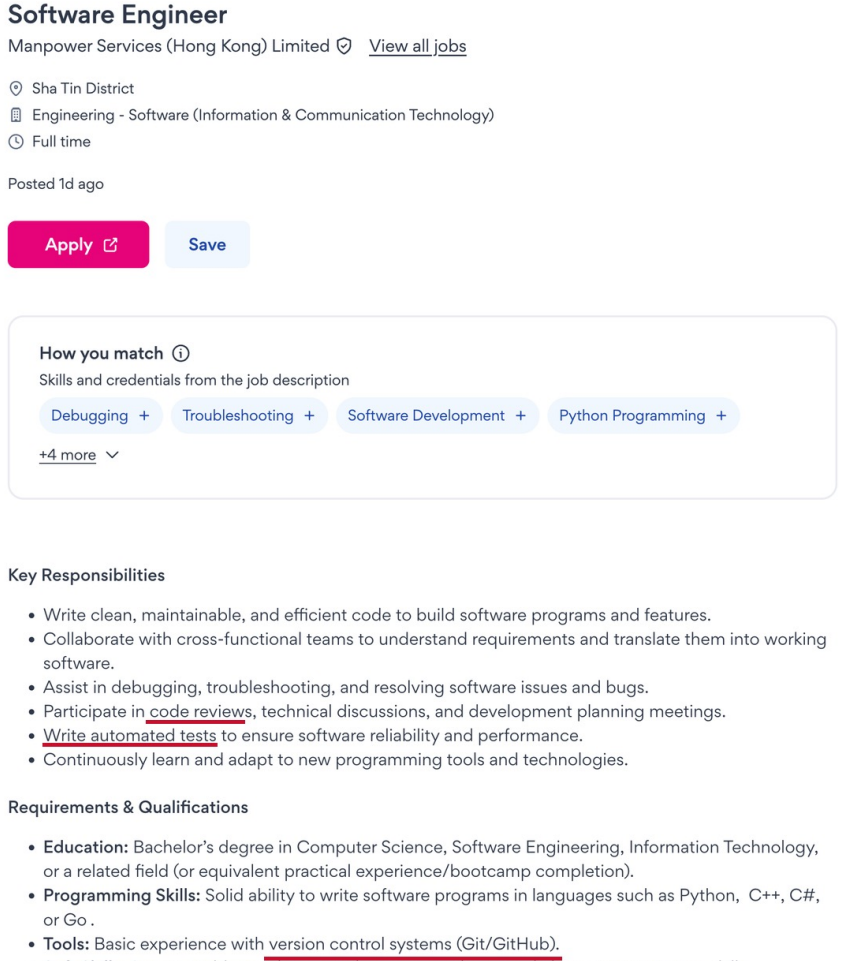
Responsibilities:

- Design and program access control systems
- Perform system testing, implementation, and system documentation
- Provide system support, training, and on-site maintenance to clients

Requirements:

- Diploma or above in Computer Science, Electrical/Electronic Engineering, or related disciplines
- Familiar with the software development life cycle (SDLC) is an advantage
- Proficiency in programming using Java, React, SQL, HTML, etc., is an advantage
- Fresh graduates are welcomed

Software Engineer - Sunevision Super E-Technology Services Ltd ("Fresh graduates are welcomed") - hk.jobsdb.com, Aug 2026



Software Engineer

Manpower Services (Hong Kong) Limited View all jobs

Sha Tin District
Engineering - Software (Information & Communication Technology)
Full time

Posted 1d ago

Apply Save

How you match
Skills and credentials from the job description
Debugging + Troubleshooting + Software Development + Python Programming +
+4 more

Key Responsibilities

- Write clean, maintainable, and efficient code to build software programs and features.
- Collaborate with cross-functional teams to understand requirements and translate them into working software.
- Assist in debugging, troubleshooting, and resolving software issues and bugs.
- Participate in code reviews, technical discussions, and development planning meetings.
- Write automated tests to ensure software reliability and performance.
- Continuously learn and adapt to new programming tools and technologies.

Requirements & Qualifications

- **Education:** Bachelor's degree in Computer Science, Software Engineering, Information Technology, or a related field (or equivalent practical experience/bootcamp completion).
- **Programming Skills:** Solid ability to write software programs in languages such as Python, C++, C#, or Go .
- **Tools:** Basic experience with version control systems (Git/GitHub).
- **Soft Skills:** Strong problem-solving mindset, eager to learn, and clear communication skills.

Software Engineer - Manpower Services (Hong Kong) Ltd - hk.jobsdb.com, Aug 2026

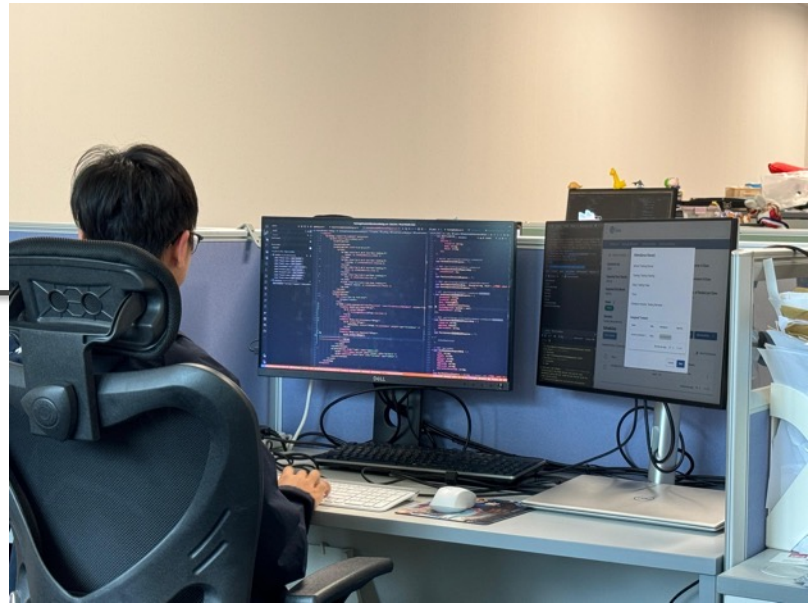
The challenges of industrial internship, final year project and the first job

Leo’s mistakes in his first internship

Before enrolling in a software engineering course, Leo worked as a web developer intern, where he encountered several challenges and made mistakes:

- **Database crash** in the first week
  - Insufficient basic technical knowledge
- **Failed client demo**
  - Low-quality work
  - Lack of testing on different devices (Android & Apple)
- **Resigned immediately** after product launch for personal vacation
  - Engaged in selfish and immoral behavior that could impact professional reputation
- Exchanged work tasks with a colleague **without notifying the manager**
  - Lack of system reliability and maintainability (from the manager's perspective)

What does my company's intern think after he completes the internship?



Intern's Evaluation Report

Q: What can you learn from this internship?

I learned how to collaborate in coding tasks. More importantly, I also learned the importance of clear communication in collaborative work as well as how to communicate effectively.

Q: How would you apply the things you have learnt in the period in your future life?

I would apply the communication skills I learned in completing collaborative work as well as daily communication. In terms of coding, I also learned things that I could improve in terms of code quality and will improve in future tasks.

Problems faced by students in final year project

- Group selection
- No clarity
- Ideal proposal
- Selection of tools/libraries
- Communication gap
- Work delay
- Risk management
- ***50-100 pages report (document)**

(Reference: David Watson, 2018)

Open Floor - Tell Us Your Story

Question: Have you faced any technical or project management related challenges in past group projects?

We will find solutions together in this course!



Why learning software engineering?

- ✧ Complete the **final year project** professionally in a team
- ✧ Prepare for the **internship** and the **first job**
- ✧ Become a **professional software engineer**
- ✧ Receive **high-paid job offers** from top tier companies
- ✧ *Perhaps even build a successful tech startup as well :)*